Permanova new

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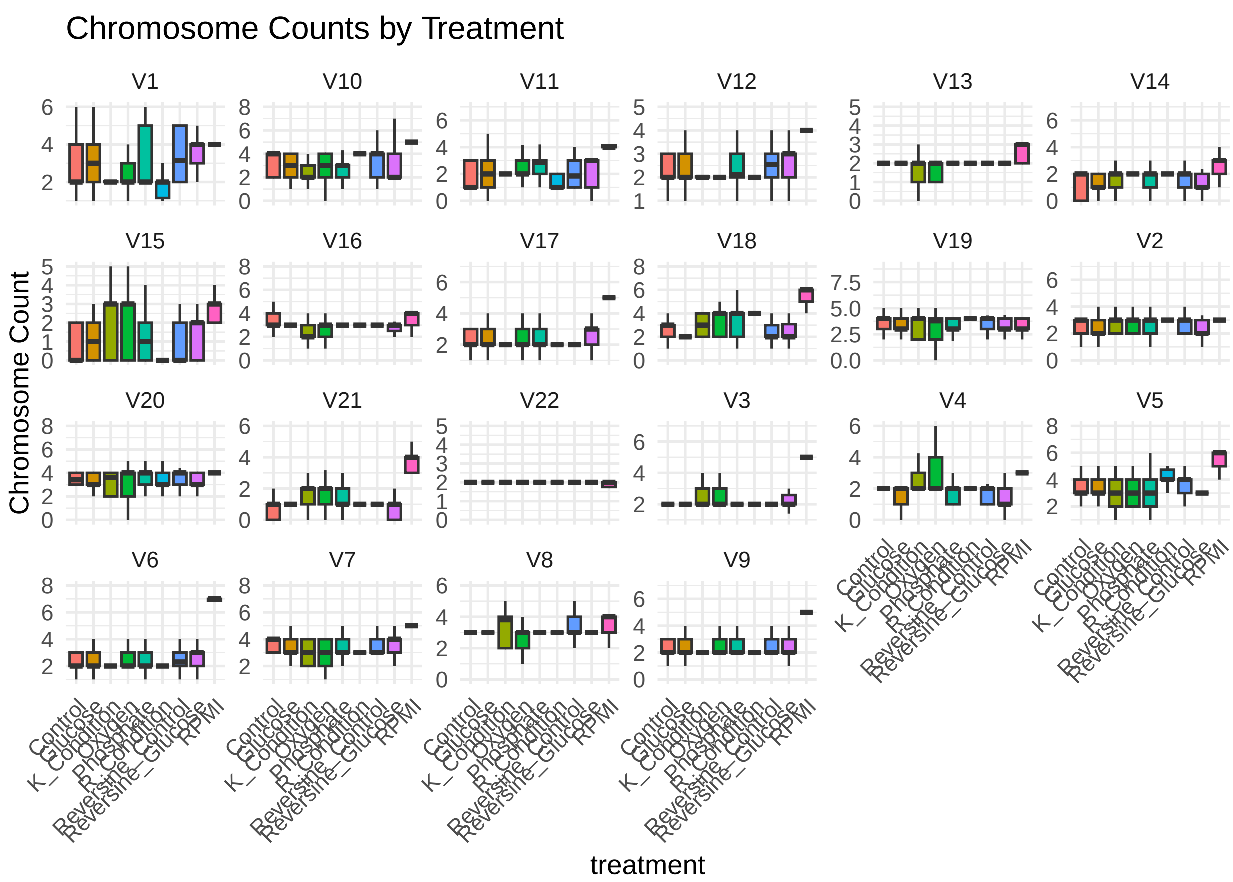
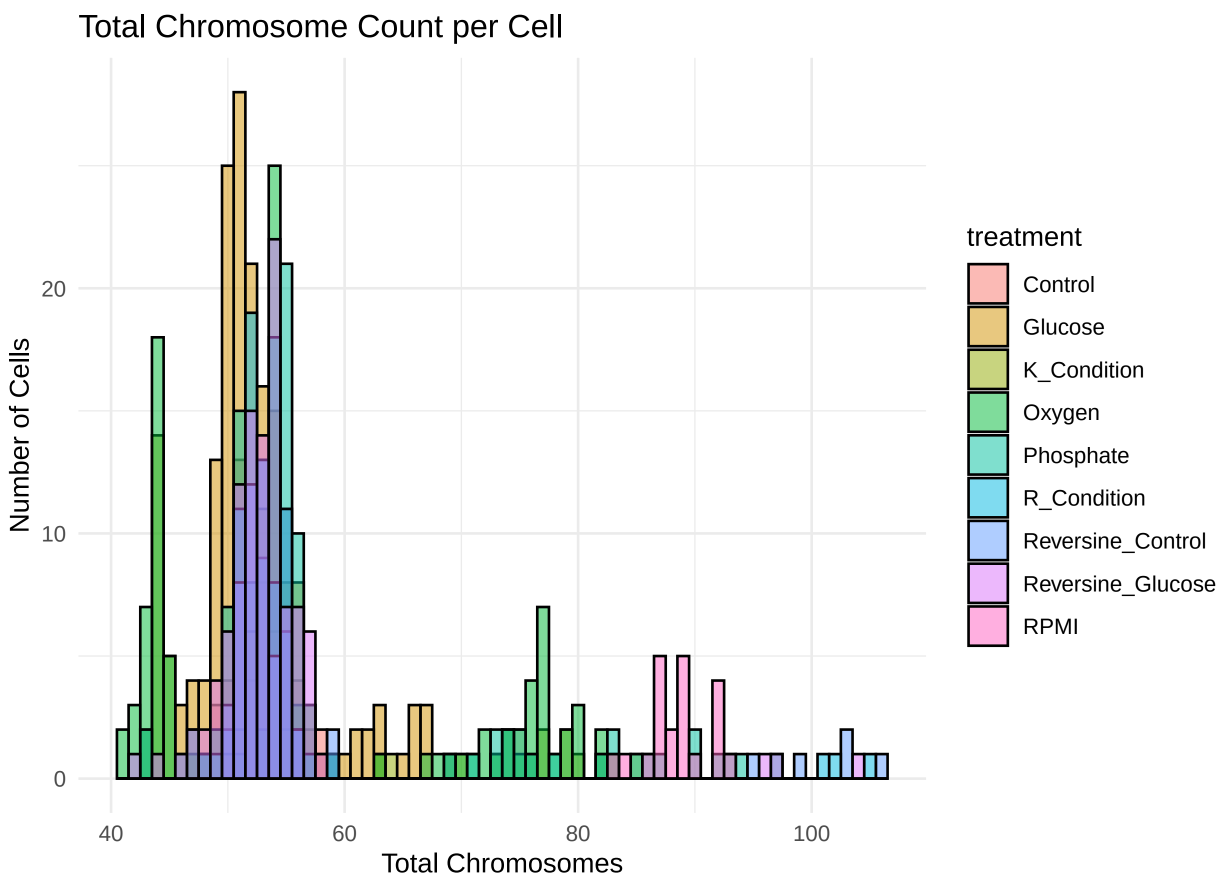
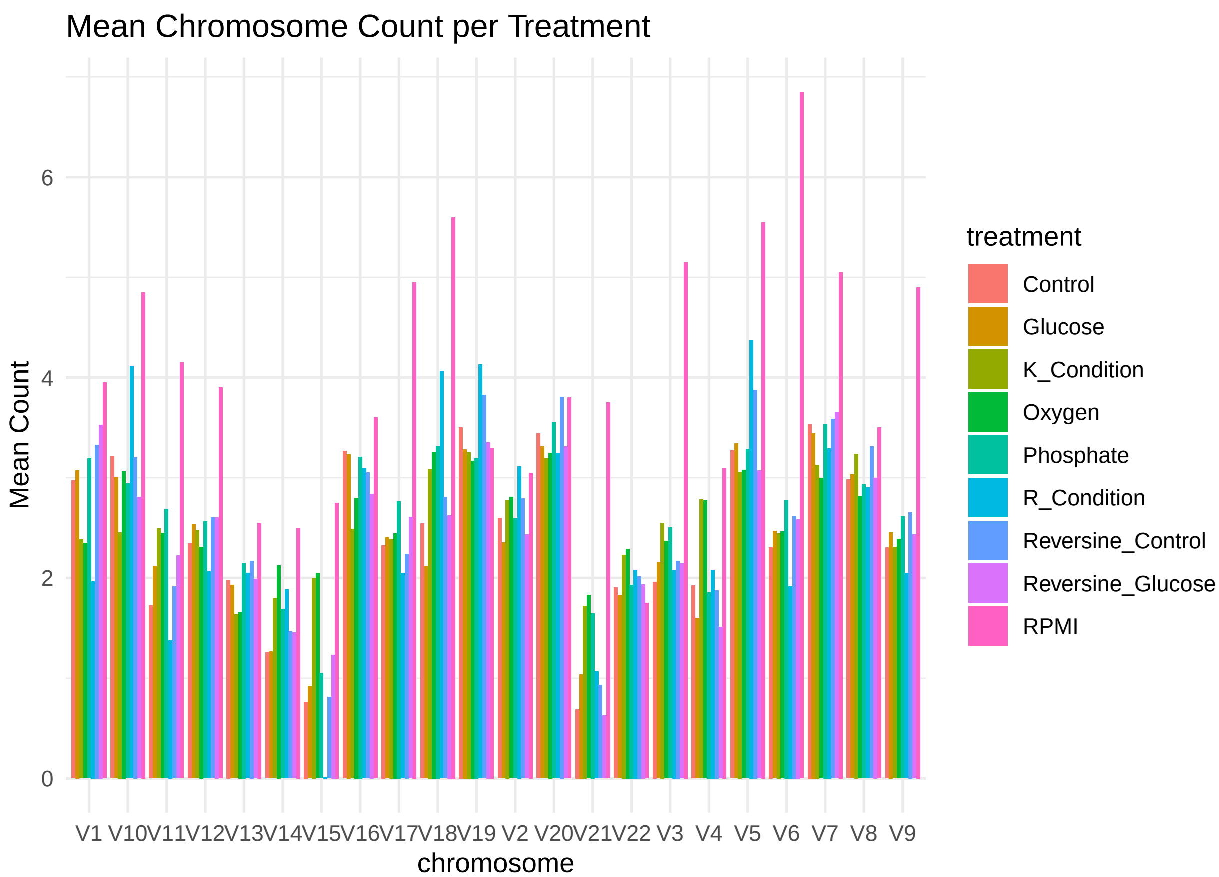
Creating a new cell line for SUM to seperate (2N and 4N)

Grouping similar treatment conditions into biologically meaningful categories:

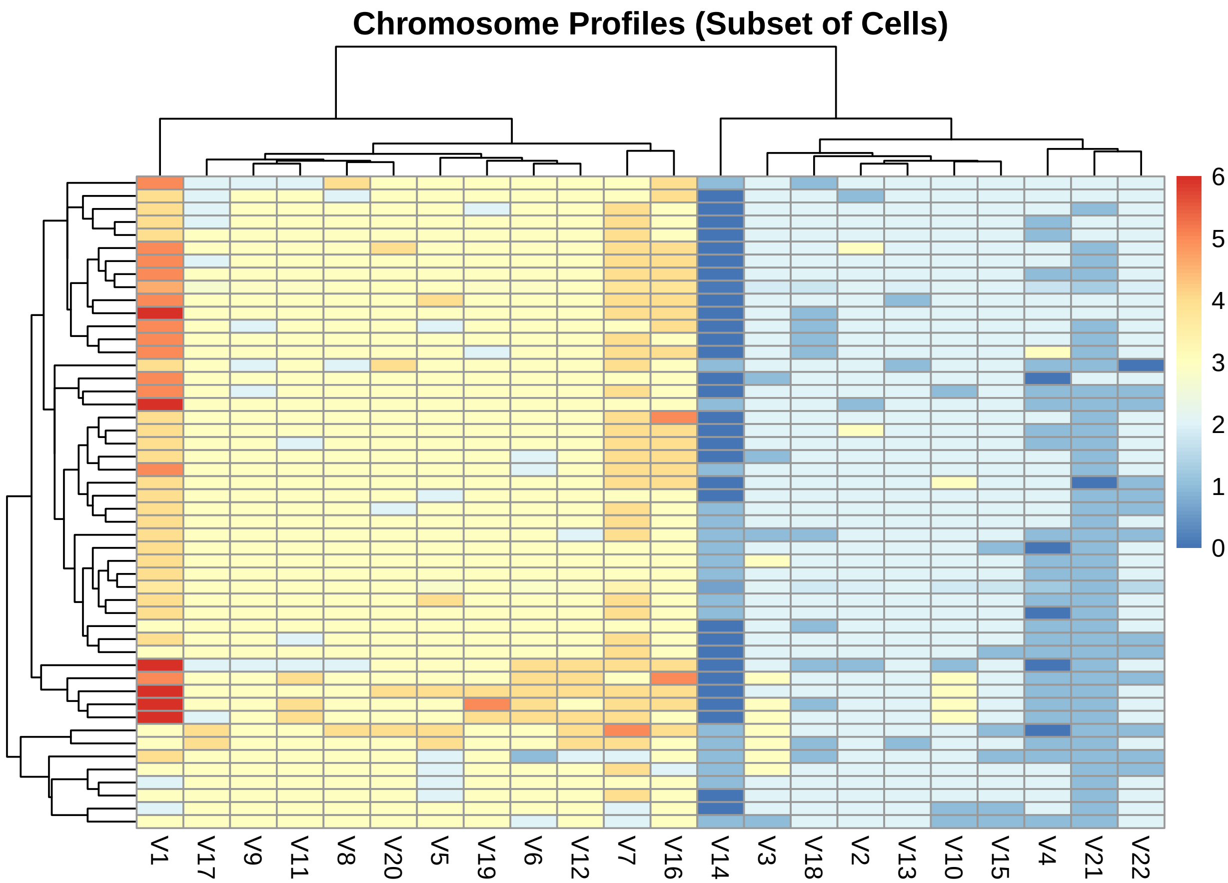




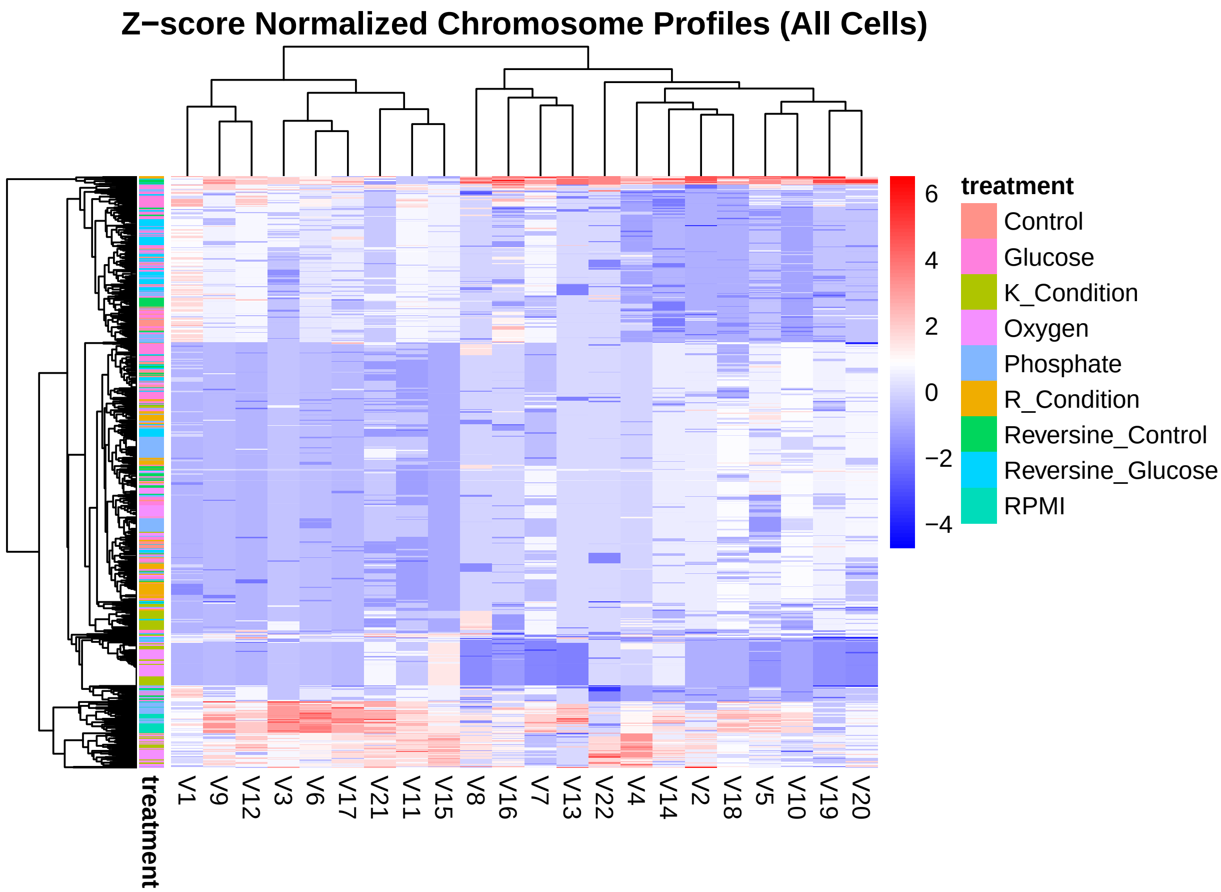

```r
# Mean per chromosome per treatment
mean_chr <- karyo_long %>%
  group_by(treatment, chromosome) %>%
  summarise(mean_count = mean(count, na.rm = TRUE), .groups = "drop")

ggplot(mean_chr, aes(x = chromosome, y = mean_count, fill = treatment)) +
  geom_bar(stat = "identity", position = "dodge") +
  theme_minimal() +
  labs(title = "Mean Chromosome Count per Treatment",
       y = "Mean Count")
```



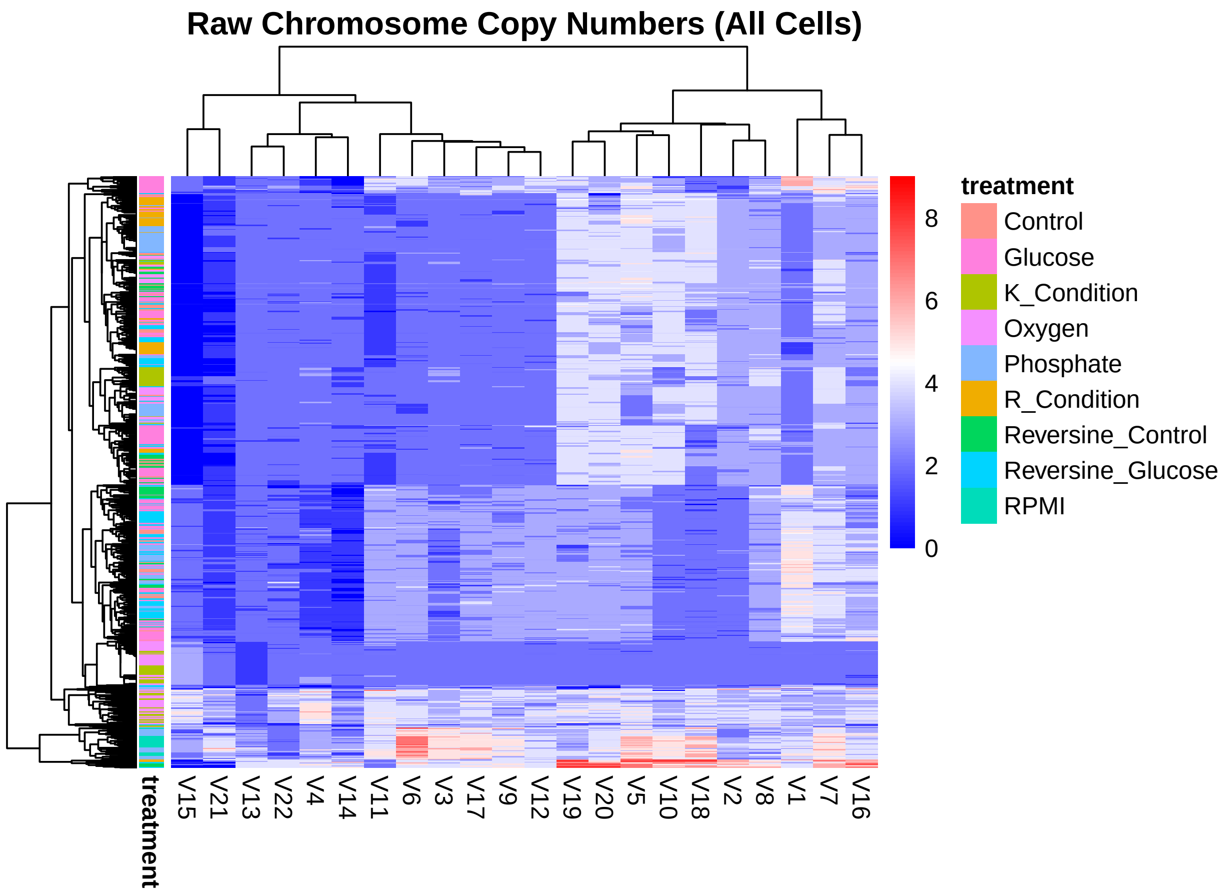

```r
library(pheatmap)

# Make a cell-by-chromosome matrix
karyo_mat <- as.matrix(karyo_df[, paste0("V", 1:22)])
rownames(karyo_mat) <- karyo_df$cloneID
# Build an annotation data frame for treatment
annotation_row <- data.frame(treatment = karyo_df$treatment)
rownames(annotation_row) <- karyo_df$cloneID


# Simple heatmap
pheatmap(karyo_mat[1:50, ], cluster_rows = TRUE, cluster_cols = TRUE,
         main = "Chromosome Profiles (Subset of Cells)",
         show_rownames = FALSE)
```




```r
library(pheatmap)

# Create matrix
karyo_mat <- as.matrix(karyo_df[, paste0("V", 1:22)])
rownames(karyo_mat) <- karyo_df$cloneID

# Create annotation
annotation_row <- data.frame(treatment = karyo_df$treatment)
rownames(annotation_row) <- karyo_df$cloneID

# Z-score normalize by chromosome (column)
karyo_mat_z <- scale(karyo_mat)

# Check dimensions
stopifnot(all(rownames(karyo_mat_z) == rownames(annotation_row)))

# Plot heatmap
pheatmap(karyo_mat_z,
         cluster_rows = TRUE,
         cluster_cols = TRUE,
         annotation_row = annotation_row,
         main = "Z-score Normalized Chromosome Profiles (All Cells)",
         show_rownames = FALSE,
         color = colorRampPalette(c("blue", "white", "red"))(100))
```




```r
# Make matrix of chromosome counts
karyo_mat <- as.matrix(karyo_df[, paste0("V", 1:22)])
rownames(karyo_mat) <- karyo_df$cloneID

# Create annotation for treatments
annotation_row <- data.frame(treatment = karyo_df$treatment)
rownames(annotation_row) <- karyo_df$cloneID

# Sanity check alignment
stopifnot(all(rownames(karyo_mat) == rownames(annotation_row)))

# Plot heatmap without normalization
pheatmap(karyo_mat,
         cluster_rows = TRUE,
         cluster_cols = TRUE,
         annotation_row = annotation_row,
         main = "Raw Chromosome Copy Numbers (All Cells)",
         show_rownames = FALSE,
         color = colorRampPalette(c("blue", "white", "red"))(100))
```



- **Note:** Cell-Level PERMANOVA without Collapsing Replicates because:

1. I am studying how treatment affects individual cells' karyotypes.
2. Each row represents a unique cell, preserving biological variability within flasks.
3. I stratify by line, so that permutations are done within each cell line.
4. So that I can retain statistical power, because you are using all the data (not just averages).

```
# Load necessary libraries
library(vegan)
library(dplyr)
library(ggplot2)

# Step 1: Make sure your karyo_df and meta_flask align
# Let's say rownames were not preserved, rebuild them together

# Create distance matrix from a clean matrix
karyo_mat <- as.matrix(karyo_df[ , 1:22])
rownames(karyo_mat) <- karyo_df$replicate_id   # Set replicate_id as rownames

# Create metadata with matching replicate_id
meta_flask <- karyo_df[, c("replicate_id", "line", "treatment")]
rownames(meta_flask) <- seq_len(nrow(meta_flask))
rownames(karyo_mat) <- seq_len(nrow(karyo_mat))
```

```r
# Now subset meta_flask to match the distance matrix
meta_flask <- meta_flask[rownames(karyo_mat), ]

# Recreate distance matrix and run adonis2
dist_flask <- dist(karyo_mat)

# Check alignment
stopifnot(all(rownames(meta_flask) == attr(dist_flask, "Labels")))

# Run adonis2 and save result
adonis_res <- adonis2(
  formula = dist_flask ~ line + treatment,
  data = meta_flask,
  strata = meta_flask$line,
  permutations = 999
)

# Convert to data frame for export
adonis_df <- as.data.frame(adonis_res)

# Clean row names as a column
adonis_df <- tibble::rownames_to_column(adonis_df, "Effect")

# Print or view
print(adonis_df)
```

```
##     Effect  Df  SumOfSqs        R2        F Pr(>F)
## 1    Model  12  9337.419 0.6338704 109.9358  0.001
## 2 Residual 762  5393.383 0.3661296       NA     NA
## 3    Total 774 14730.802 1.0000000       NA     NA
```

Interpretation: A PERMANOVA was conducted to assess whether differences in karyotype profiles among individual cells were significantly associated with cell line and treatment conditions, while accounting for stratified permutation within each cell line. The overall model was highly significant (F = 109.94, p = 0.001), indicating that the combined effects of cell line and treatment explain approximately 63.4% of the total variation in the chromosome features ($R^2 = 0.634$). This suggests that both the intrinsic characteristics of different cell lines and the external interventions from treatments contribute meaningfully to chromosomal differences observed in the data. The remaining 36.6% of the variation is unexplained by these variables and may reflect biological heterogeneity, noise, or other unmeasured factors.

```r
# Step 6: Test if treatment affects karyotype variability (dispersion)
library(vegan)

# Step 1: Compute dispersion
dispersion <- betadisper(dist_flask, group = meta_flask$treatment)

# Step 2: Run ANOVA
dispersion_anova <- anova(dispersion)

# Step 3: Convert to tidy table
dispersion_df <- broom::tidy(dispersion_anova)

# Optional: Clean row names if needed
```

```r
dispersion_df <- tibble::rownames_to_column(dispersion_df, var = "Effect")

# View or print
print(dispersion_df)
```

```
## # A tibble: 2 x 7
##   Effect term         df sumsq meansq statistic   p.value
##   <chr>  <chr>     <int> <dbl>  <dbl>     <dbl>     <dbl>
## 1 1      Groups        8  273.  34.1       10.8  2.01e-14
## 2 2      Residuals   766 2428.   3.17      NA   NA
```

Note: We discussed about using variance. This PERMISP is meant to check the asumption of PERMANOV and also Multivariate dispersion (spread) across groups. it test Overall karyotype variability across treatments.

Intepretation: The analysis of multivariate dispersions (PERMDISP) revealed a significant difference in dispersion among treatment groups ($F = 10.76$, $p < 0.001$), suggesting that variability within treatments is not uniform. This violates a key assumption of PERMANOVA, which presumes homogeneous dispersions. Consequently, although significant PERMANOVA results may indicate group differences, part of this difference could stem from heterogeneous group variances (dispersion), rather than true differences in centroid locations. Therefore, any inference about treatment effects on karyotype composition should be interpreted with caution and supported with additional visualization (e.g., NMDS/CAPscale plots) or models robust to dispersion differences, such as generalized linear models.

```r
permutest(dispersion, permutations = 999, pairwise = T ) ->t
t$pairwise %>% kableExtra::kable(digits = 4, caption = "Pairwise Permuted P-Values for Group Dispersion
```

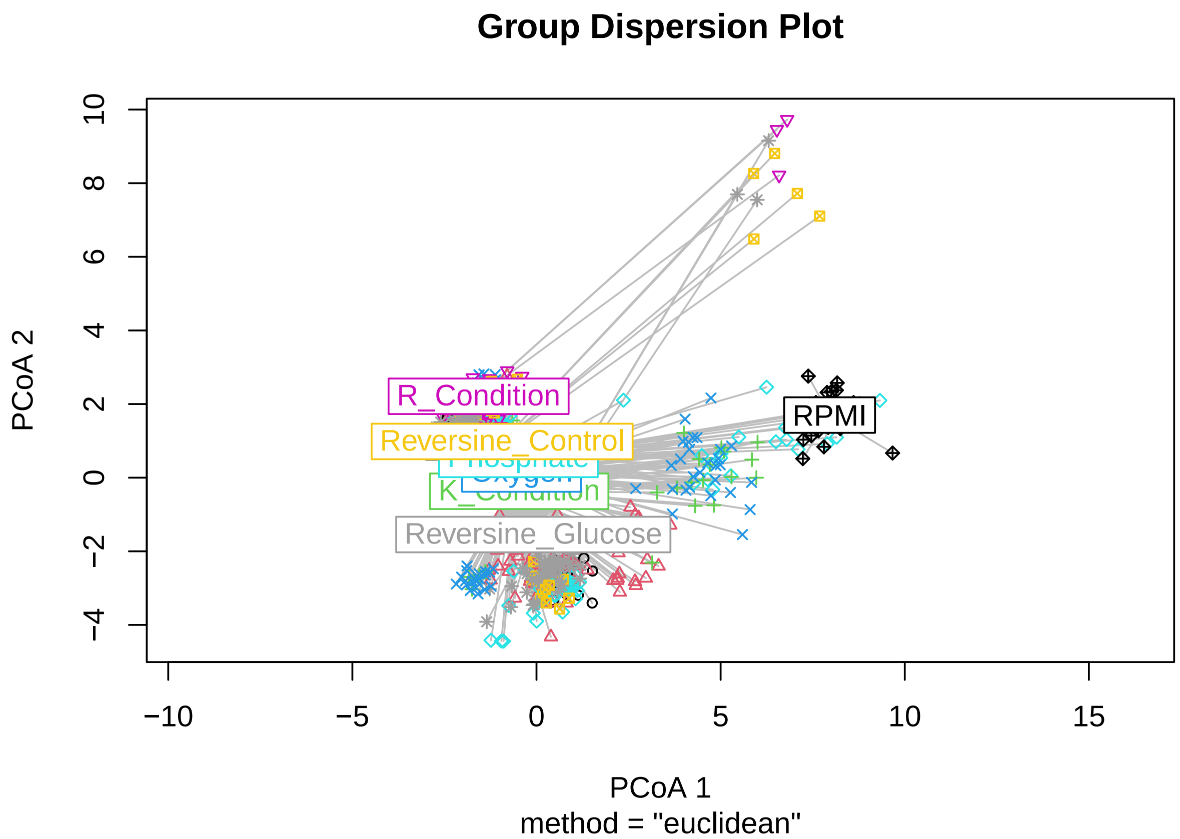
Interpretation: The pairwise PERMDISP analysis revealed significant differences in dispersion (i.e., within-group variability) among several treatment groups, indicating that the assumption of homogenous variance required by PERMANOVA is violated in many comparisons. Notably, treatments such as Glucose, Oxygen, R_Condition, and Reversine_Glucose showed significant differences in dispersion when compared with multiple other treatments ($p < 0.05$). For example, Glucose differed significantly from K_Condition, Phosphate, and R_Condition, while R_Condition differed in dispersion from nearly all other treatments, suggesting it may be especially destabilizing to karyotypic structure. In contrast, treatments like Control vs Reversine_Glucose and Reversine_Control vs Reversine_Glucose did not differ significantly in their dispersion, indicating comparable within-group variability. These findings imply that differences observed in PERMANOVA results could, at least in part, be driven by differences in dispersion rather than shifts in group centroids alone.

```r
plot(dispersion, hull = FALSE, main = "Group Dispersion Plot")
```

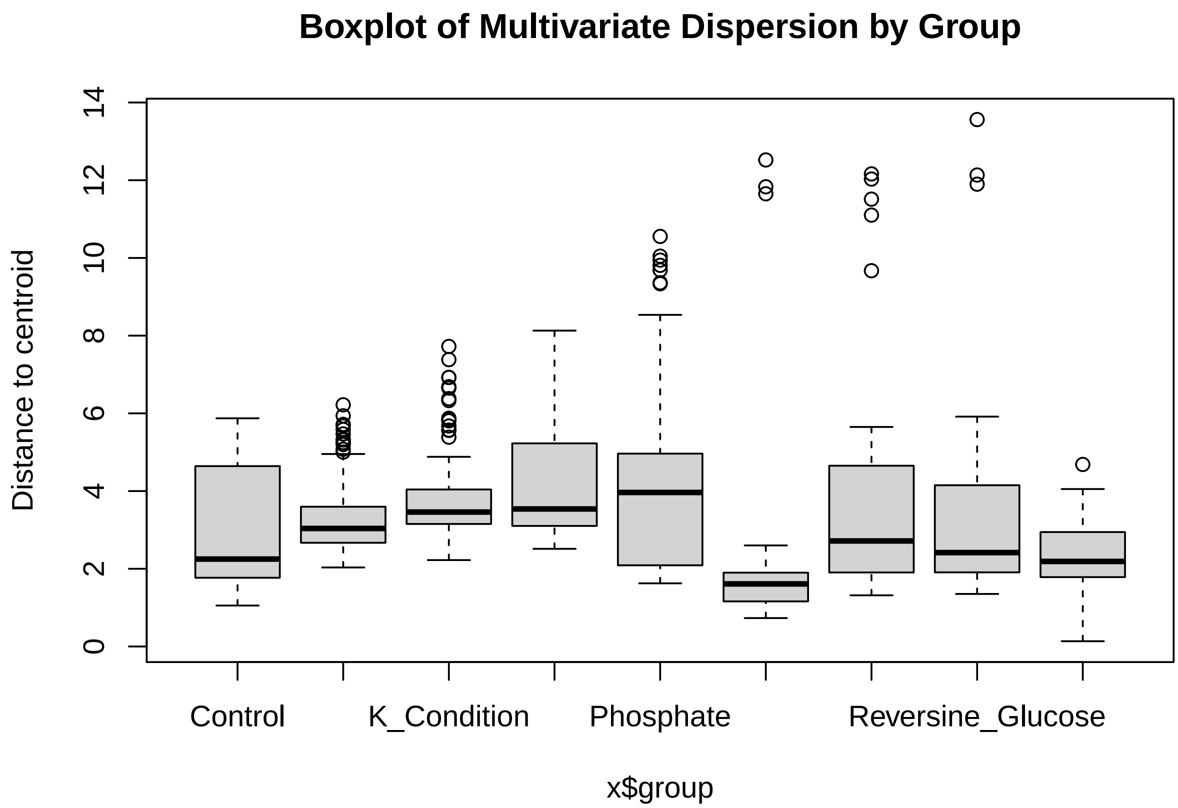
Table 1: Pairwise Permuted P-Values for Group Dispersion

	x
Control-Glucose	0.1406
Control-K_Condition	0.0014
Control-Oxygen	0.0000
Control-Phosphate	0.0072
Control-R_Condition	0.0066
Control-Reversine_Control	0.0642
Control-Reversine_Glucose	0.5428
Control-RPMI	0.0674
Glucose-K_Condition	0.0001
Glucose-Oxygen	0.0000
Glucose-Phosphate	0.0006
Glucose-R_Condition	0.0000
Glucose-Reversine_Control	0.0277
Glucose-Reversine_Glucose	0.7357
Glucose-RPMI	0.0000
K_Condition-Oxygen	0.3735
K_Condition-Phosphate	0.8504
K_Condition-R_Condition	0.0000
K_Condition-Reversine_Control	0.7889
K_Condition-Reversine_Glucose	0.0153
K_Condition-RPMI	0.0000
Oxygen-Phosphate	0.6052
Oxygen-R_Condition	0.0000
Oxygen-Reversine_Control	0.3534
Oxygen-Reversine_Glucose	0.0002
Oxygen-RPMI	0.0000
Phosphate-R_Condition	0.0000
Phosphate-Reversine_Control	0.6810
Phosphate-Reversine_Glucose	0.0112
Phosphate-RPMI	0.0021
R_Condition-Reversine_Control	0.0001
R_Condition-Reversine_Glucose	0.0005
R_Condition-RPMI	0.5241
Reversine_Control-Reversine_Glucose	0.1150
Reversine_Control-RPMI	0.0178
Reversine_Glucose-RPMI	0.0554

	x
Control-Glucose	0.144
Control-K_Condition	0.002
Control-Oxygen	0.001
Control-Phosphate	0.013
Control-R_Condition	0.007
Control-Reversine_Control	0.080
Control-Reversine_Glucose	0.517
Control-RPMI	0.063
Glucose-K_Condition	0.001
Glucose-Oxygen	0.001
Glucose-Phosphate	0.001
Glucose-R_Condition	0.001
Glucose-Reversine_Control	0.031
Glucose-Reversine_Glucose	0.708
Glucose-RPMI	0.001
K_Condition-Oxygen	0.379
K_Condition-Phosphate	0.854
K_Condition-R_Condition	0.001
K_Condition-Reversine_Control	0.800
K_Condition-Reversine_Glucose	0.019
K_Condition-RPMI	0.001
Oxygen-Phosphate	0.602
Oxygen-R_Condition	0.001
Oxygen-Reversine_Control	0.360
Oxygen-Reversine_Glucose	0.001
Oxygen-RPMI	0.001
Phosphate-R_Condition	0.001
Phosphate-Reversine_Control	0.709
Phosphate-Reversine_Glucose	0.011
Phosphate-RPMI	0.003
R_Condition-Reversine_Control	0.001
R_Condition-Reversine_Glucose	0.001
R_Condition-RPMI	0.528
Reversine_Control-Reversine_Glucose	0.119
Reversine_Control-RPMI	0.021
Reversine_Glucose-RPMI	0.050




```r
boxplot(dispersion, main = "Boxplot of Multivariate Dispersion by Group")
```



- 2: GLMM for Per-Chromosome Changes

```
library(lme4)

# Binary outcome: gain defined as >2 copies
karyo_df$gain_chr08 <- ifelse(karyo_df$V8 > 2, 1, 0)

# GLMM: treatment as fixed effect, replicate as random effect
model_chr8 <- glmer(gain_chr08 ~ treatment + (1 | replicate_id),
                    data = karyo_df, family = binomial)

summary(model_chr8)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: gain_chr08 ~ treatment + (1 | replicate_id)
##    Data: karyo_df
##
##      AIC       BIC    logLik -2*log(L)  df.resid
##    452.8     499.4    -216.4     432.8       765
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -4.9985  0.1067  0.2001  0.3134  0.7307
##
## Random effects:
##  Groups       Name        Variance Std.Dev.
##  replicate_id (Intercept) 4.799    2.191
## Number of obs: 775, groups:  replicate_id, 41
##
## Fixed effects:
##                             Estimate Std. Error z value Pr(>|z|)
## (Intercept)                   4.1388     1.5836   2.614  0.00896 **
## treatmentGlucose             -0.6897     1.8175  -0.379  0.70433
## treatmentK_Condition         -1.1702     2.0723  -0.565  0.57228
## treatmentOxygen              -2.6862     1.9029  -1.412  0.15806
## treatmentPhosphate           -1.6484     1.8526  -0.890  0.37360
## treatmentR_Condition         -1.8697     2.1039  -0.889  0.37418
## treatmentReversine_Control  -0.3749     2.1819  -0.172  0.86356
## treatmentReversine_Glucose  -1.4482     1.8696  -0.775  0.43857
## treatmentRPMI                -1.6832     2.8224  -0.596  0.55093
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) trtmnG trtK_C trtmnO trtmnP trtmntR_Cn trtmntRv_C trtR_G
## tretmntGlcs -0.861
## trtmntK_Cnd -0.743  0.660
## trtmntOxygn -0.826  0.717  0.626
## trtmntPhsph -0.853  0.736  0.637  0.705
## trtmntR_Cnd -0.745  0.649  0.569  0.620  0.637
## trtmntRvr_C -0.718  0.626  0.549  0.598  0.614  0.540
## trtmntRvr_G -0.842  0.730  0.636  0.698  0.719  0.631      0.608
## tretmntRPMI -0.561  0.483  0.417  0.463  0.478  0.418      0.403      0.472
```

Note: why use chr08 (This is reference to the fact that biological relevance, in many cancer gain of chr08 are commonly associated with tumor progression). **Interpretation:** The generalized linear mixed model assessing the impact of treatment on chromosome 08 gain revealed that no treatment condition had a statistically significant effect compared to the control. The control group showed a significantly high baseline probability of chr08 gain. Variability among replicate flasks was substantial, supporting the inclusion of random effects in the model.

P-Adjustment (e.g., Bonferroni and Benjamini-Hochberg)

```r
# Extract p-values from model summary
p_vals <- summary(model_chr8)$coefficients[-1, "Pr(>|z|)"]  # Exclude Intercept

# Adjust p-values (Bonferroni)
p_adj_bonf <- p.adjust(p_vals, method = "bonferroni")

# Adjust p-values (Benjamini-Hochberg)
p_adj_bh <- p.adjust(p_vals, method = "BH")

# Combine results
p_table <- data.frame(
```

```r
  Treatment = names(p_vals),
  Raw_P = p_vals,
  Bonferroni = p_adj_bonf,
  BH = p_adj_bh
)

p_table %>% kableExtra::kable()
```

	Treatment	Raw_P	Bonferroni	BH
treatmentGlucose	treatmentGlucose	0.7043349	1	0.8049542
treatmentK_Condition	treatmentK_Condition	0.5722773	1	0.7630364
treatmentOxygen	treatmentOxygen	0.1580622	1	0.7630364
treatmentPhosphate	treatmentPhosphate	0.3736007	1	0.7630364
treatmentR_Condition	treatmentR_Condition	0.3741779	1	0.7630364
treatmentReversine_Control	treatmentReversine_Control	0.8635605	1	0.8635605
treatmentReversine_Glucose	treatmentReversine_Glucose	0.4385716	1	0.7630364
treatmentRPMI	treatmentRPMI	0.5509301	1	0.7630364

Interpretation: A generalized linear mixed model (GLMM) was used to assess whether different treatment conditions significantly influenced the likelihood of gain of chromosome 08 across replicate cell populations. None of the treatments showed a statistically significant effect, either before or after adjusting for multiple testing using Bonferroni or Benjamini-Hochberg (BH) procedures. The lowest raw p-value was observed for the Oxygen treatment ($p = 0.158$), but this did not remain significant after adjustment. These findings suggest that gain of chromosome 08 is not significantly associated with any specific treatment condition in this dataset, implying that observed variations are likely due to random chance or other uncontrolled biological factors.

1. Repeat the GLMM for other chromosomes (chr07, chr13, etc.)

```r
library(lme4)
library(dplyr)
library(tidyr)

# Example for chr07 and chr13
karyo_df <- karyo_df %>%
  mutate(
    gain_chr07 = ifelse(V7 > 2, 1, 0),
    gain_chr13 = ifelse(V13 > 2, 1, 0)
  )

# Fit GLMMs for chr07 and chr13
fit_chr07 <- glmer(gain_chr07 ~ treatment + (1 | replicate_id),
                   data = karyo_df, family = binomial)
fit_chr13 <- glmer(gain_chr13 ~ treatment + (1 | replicate_id),
                   data = karyo_df, family = binomial)

# Summarize models
summary(fit_chr07)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
```

```
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: gain_chr07 ~ treatment + (1 | replicate_id)
##    Data: karyo_df
##
##      AIC       BIC    logLik -2*log(L)  df.resid
##    246.6     293.2    -113.3     226.6       765
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -5.1160  0.0000  0.0657  0.2019  0.3827
##
## Random effects:
##  Groups       Name        Variance Std.Dev.
##  replicate_id (Intercept) 9.135    3.022
## Number of obs: 775, groups:  replicate_id, 41
##
## Fixed effects:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)                  23.093  13943.493   0.002    0.999
## treatmentGlucose            -18.271  13943.493  -0.001    0.999
## treatmentK_Condition        -20.365  13943.493  -0.001    0.999
## treatmentOxygen             -21.840  13943.493  -0.002    0.999
## treatmentPhosphate          -17.417  13943.493  -0.001    0.999
## treatmentR_Condition        -18.484  13943.493  -0.001    0.999
## treatmentReversine_Control   -2.520  14440.640   0.000    1.000
## treatmentReversine_Glucose  -18.470  13943.493  -0.001    0.999
## treatmentRPMI                -4.743  14101.729   0.000    1.000
##
## Correlation of Fixed Effects:
##             (Intr) trtmnG trtK_C trtmnO trtmnP trtmntR_Cn trtmntRv_C trtR_G
## tretmntGlcs -1.000
## trtmntK_Cnd -1.000  1.000
## trtmntOxygn -1.000  1.000  1.000
## trtmntPhsph -1.000  1.000  1.000  1.000
## trtmntR_Cnd -1.000  1.000  1.000  1.000  1.000
## trtmntRvr_C -0.966  0.966  0.966  0.966  0.966  0.966
## trtmntRvr_G -1.000  1.000  1.000  1.000  1.000  1.000      0.966
## tretmntRPMI -0.989  0.989  0.989  0.989  0.989  0.989      0.955      0.989
## optimizer (Nelder_Mead) convergence code: 0 (OK)
## unable to evaluate scaled gradient
## Model failed to converge: degenerate  Hessian with 2 negative eigenvalues
```

```
summary(fit_chr13)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: gain_chr13 ~ treatment + (1 | replicate_id)
##    Data: karyo_df
##
##      AIC       BIC    logLik -2*log(L)  df.resid
##    251.5     298.0    -115.7     231.5       765
##
```

```
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.0932 -0.0807 -0.0489 -0.0345  5.3000
##
## Random effects:
##  Groups       Name        Variance Std.Dev.
##  replicate_id (Intercept) 6.312    2.512
## Number of obs: 775, groups:  replicate_id, 41
##
## Fixed effects:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)                  -19.38     127.02  -0.153    0.879
## treatmentGlucose              12.80     127.02   0.101    0.920
## treatmentK_Condition          13.42     127.02   0.106    0.916
## treatmentOxygen               13.31     127.03   0.105    0.917
## treatmentPhosphate            15.16     127.02   0.119    0.905
## treatmentR_Condition          14.78     127.03   0.116    0.907
## treatmentReversine_Control    16.47     127.03   0.130    0.897
## treatmentReversine_Glucose    14.47     127.03   0.114    0.909
## treatmentRPMI                 19.68     127.04   0.155    0.877
##
## Correlation of Fixed Effects:
##             (Intr) trtmnG trtK_C trtmnO trtmnP trtmntR_Cn trtmntRv_C trtR_G
## tretmntGlcs -1.000
## trtmntK_Cnd -1.000  1.000
## trtmntOxygn -1.000  1.000  1.000
## trtmntPhsph -1.000  1.000  1.000  1.000
## trtmntR_Cnd -1.000  1.000  1.000  1.000  1.000
## trtmntRvr_C -1.000  1.000  1.000  1.000  1.000  1.000
## trtmntRvr_G -1.000  1.000  1.000  1.000  1.000  1.000      1.000
## tretmntRPMI -1.000  1.000  1.000  1.000  1.000  1.000      1.000      1.000
## optimizer (Nelder_Mead) convergence code: 0 (OK)
## Model is nearly unidentifiable: large eigenvalue ratio
##  - Rescale variables?
```

```
for (cell_line in unique(karyo_df$line)) {
  cat("\n\n### Cell Line:", cell_line, "###\n")
  line_df <- subset(karyo_df, line == cell_line)
  line_mat <- vegdist(as.matrix(line_df[, 1:22]), method = "manhattan")

  print(adonis2(line_mat ~ treatment,
                data = line_df,
                permutations = 999))
}
```

```
##
##
## ### Cell Line: HGC-27 ###
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##           Df SumOfSqs      R2      F Pr(>F)
```

```
## Model      4   1578.3 0.12227 7.7661  0.001 ***
## Residual 223  11330.0 0.87773
## Total    227  12908.3 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## ### Cell Line: MDAMB231 ###
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##          Df SumOfSqs      R2      F Pr(>F)
## Model     1    999.3 0.19734 9.8344  0.001 ***
## Residual 40   4064.6 0.80266
## Total    41   5063.9 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## ### Cell Line: SNU-668 ###
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##           Df SumOfSqs      R2      F Pr(>F)
## Model      7     6470 0.16122 10.736  0.001 ***
## Residual 391    33663 0.83878
## Total    398    40134 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## ### Cell Line: SUM-159 ###
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##          Df SumOfSqs      R2     F Pr(>F)
## Model     1     2.03 0.00071 0.041   0.85
## Residual 58  2874.83 0.99929
## Total    59  2876.86 1.00000
##
##
## ### Cell Line: SUM-159-4N ###
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##          Df SumOfSqs      R2      F Pr(>F)
```

```
## Model     1    167.5 0.02961 1.3426  0.196
## Residual 44   5490.6 0.97039
## Total    45   5658.2 1.00000
```

```r
library(vegan)
library(pairwiseAdonis)

library(vegan)

library(vegan)
library(dplyr)

# Initialize a list to hold pairwise results for each cell line
all_pairwise_results <- list()

for (cell_line in unique(karyo_df$line)) {
  cat("\n\n### Cell Line:", cell_line, "###\n")

  # Subset data for the current cell line
  line_df <- subset(karyo_df, line == cell_line)
  treatments <- unique(line_df$treatment)

  # Compute overall distance matrix
  line_mat <- vegdist(as.matrix(line_df[, 1:22]), method = "manhattan")

  # Run overall PERMANOVA
  cat("Overall PERMANOVA:\n")
  print(adonis2(line_mat ~ treatment, data = line_df, permutations = 999))

  # Pairwise comparisons (manual)
  pairwise_results <- data.frame()

  for (i in 1:(length(treatments) - 1)) {
    for (j in (i + 1):length(treatments)) {
      t1 <- treatments[i]
      t2 <- treatments[j]

      # Subset for two treatment groups
      sub_df <- line_df[line_df$treatment %in% c(t1, t2), ]

      # Check for replicates
      if (sum(sub_df$treatment == t1) < 2 || sum(sub_df$treatment == t2) < 2) next

      # Compute distance
      sub_mat <- vegdist(as.matrix(sub_df[, 1:22]), method = "manhattan")

      # Run pairwise PERMANOVA
      result <- adonis2(sub_mat ~ treatment, data = sub_df, permutations = 999)

      # Collect results
      pairwise_results <- rbind(pairwise_results, data.frame(
        Cell_Line = cell_line,
        Group1 = t1,
        Group2 = t2,
```

```r
        F_value = result$F[1],
        P_value = result$`Pr(>F)`[1]
      ))
    }
  }

  # Adjust p-values
  if (nrow(pairwise_results) > 0) {
    pairwise_results$P_adj_BH <- p.adjust(pairwise_results$P_value, method = "BH")
    pairwise_results$P_adj_Bonf <- p.adjust(pairwise_results$P_value, method = "bonferroni")
    all_pairwise_results[[cell_line]] <- pairwise_results
  } else {
    cat("Not enough replicates per group for pairwise comparisons.\n")
  }
}
```

```
##
##
## ### Cell Line: HGC-27 ###
## Overall PERMANOVA:
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##           Df SumOfSqs      R2      F Pr(>F)
## Model      4   1578.3 0.12227 7.7661  0.001 ***
## Residual 223  11330.0 0.87773
## Total    227  12908.3 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## ### Cell Line: MDAMB231 ###
## Overall PERMANOVA:
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##          Df SumOfSqs      R2      F Pr(>F)
## Model     1    999.3 0.19734 9.8344  0.001 ***
## Residual 40   4064.6 0.80266
## Total    41   5063.9 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## ### Cell Line: SNU-668 ###
## Overall PERMANOVA:
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
```

```
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##           Df SumOfSqs      R2      F Pr(>F)
## Model      7     6470 0.16122 10.736  0.001 ***
## Residual 391    33663 0.83878
## Total    398    40134 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## ### Cell Line: SUM-159 ###
## Overall PERMANOVA:
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##          Df SumOfSqs      R2     F Pr(>F)
## Model     1     2.03 0.00071 0.041  0.859
## Residual 58  2874.83 0.99929
## Total    59  2876.86 1.00000
##
##
## ### Cell Line: SUM-159-4N ###
## Overall PERMANOVA:
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = line_mat ~ treatment, data = line_df, permutations = 999)
##          Df SumOfSqs      R2      F Pr(>F)
## Model     1    167.5 0.02961 1.3426  0.206
## Residual 44   5490.6 0.97039
## Total    45   5658.2 1.00000
```

```r
library(vegan)
library(dplyr)

# Initialize list to store results
adonis_results <- list()

# Loop over unique cell lines
for (cell_line in unique(karyo_df$line)) {

  line_df <- subset(karyo_df, line == cell_line)
  line_mat <- vegdist(as.matrix(line_df[, 1:22]), method = "manhattan")

  # Run adonis2 and store results
  a <- adonis2(line_mat ~ treatment, data = line_df, permutations = 999)

  # Extract relevant row (usually the first row corresponds to 'treatment')
  a_row <- as.data.frame(a)[1, ]

  # Add cell line info
```

```r
  a_row$cell_line <- cell_line

  # Append to list
  adonis_results[[cell_line]] <- a_row
}

# Combine all into a data frame
adonis_table <- bind_rows(adonis_results)

# Reorder columns (optional)
adonis_table <- adonis_table %>%
  select(cell_line, everything())

# View the result
print(adonis_table)
```

```
##            cell_line Df    SumOfSqs           R2           F Pr(>F)
## Model...1     HGC-27  4 1578.296223 0.1222702851  7.76613607  0.001
## Model...2   MDAMB231  1  999.315000 0.1973409915  9.83436251  0.001
## Model...3    SNU-668  7 6470.429345 0.1612213047 10.73627823  0.001
## Model...4    SUM-159  1    2.030174 0.0007056911  0.04095899  0.844
## Model...5 SUM-159-4N  1  167.540580 0.0296103623  1.34261115  0.221
```

Interpretation: PERMANOVA analysis within individual cell lines revealed that treatment significantly influenced karyotypic variation in HGC-27 ($R^2 = 12.2\%$, $p = 0.001$), MDAMB231 ($R^2 = 19.7\%$, $p = 0.001$), and SNU-668 ($R^2 = 16.1\%$, $p = 0.001$), suggesting strong treatment effects on chromosomal composition within these lines. In contrast, no significant treatment effects were observed in SUM-159 ($R^2 = 0.07\%$, $p = 0.848$) and SUM-159-4N ($R^2 = 2.9\%$, $p = 0.229$), indicating greater karyotypic stability or treatment resistance in those lines.

```r
# Optional: combine all into one table
combined_pairwise_results <- do.call(rbind, all_pairwise_results)

# View combined results
combined_pairwise_results %>% kableExtra::kable()
```

	Cell_Line	Group1	Group2	F_value	P_value	P_adj_BH	P_adj_Bonf
HGC-27.1	HGC-27	Control	Glucose	3.863941	0.003	0.0030000	0.030
HGC-27.2	HGC-27	Control	Phosphate	9.250636	0.001	0.0011111	0.010
HGC-27.3	HGC-27	Control	Reversine_Control	13.829547	0.001	0.0011111	0.010
HGC-27.4	HGC-27	Control	Reversine_Glucose	18.685173	0.001	0.0011111	0.010
HGC-27.5	HGC-27	Glucose	Phosphate	8.574273	0.001	0.0011111	0.010
HGC-27.6	HGC-27	Glucose	Reversine_Control	6.504384	0.001	0.0011111	0.010
HGC-27.7	HGC-27	Glucose	Reversine_Glucose	6.160983	0.001	0.0011111	0.010
HGC-27.8	HGC-27	Phosphate	Reversine_Control	6.683576	0.001	0.0011111	0.010
HGC-27.9	HGC-27	Phosphate	Reversine_Glucose	8.420818	0.001	0.0011111	0.010
HGC-27.10	HGC-27	Reversine_Control	Reversine_Glucose	11.607263	0.001	0.0011111	0.010
MDAMB231	MDAMB231	Phosphate	RPMI	9.834362	0.001	0.0010000	0.001
SNU-668.1	SNU-668	Control	Glucose	8.466687	0.001	0.0014000	0.028

	Cell_Line	Group1	Group2	F_value	P_value	P_adj_BH	P_adj_Bonf
SNU-668.2	SNU-668	Control	K_Condition	48.349747	0.001	0.0014000	0.028
SNU-668.3	SNU-668	Control	Oxygen	24.726031	0.001	0.0014000	0.028
SNU-668.4	SNU-668	Control	Phosphate	41.516836	0.001	0.0014000	0.028
SNU-668.5	SNU-668	Control	R_Condition	7.578768	0.001	0.0014000	0.028
SNU-668.6	SNU-668	Control	Reversine_Control	6.072591	0.003	0.0036522	0.084
SNU-668.7	SNU-668	Control	Reversine_Glucose	3.619705	0.018	0.0201600	0.504
SNU-668.8	SNU-668	Glucose	K_Condition	61.156951	0.001	0.0014000	0.028
SNU-668.9	SNU-668	Glucose	Oxygen	24.790881	0.001	0.0014000	0.028
SNU-668.10	SNU-668	Glucose	Phosphate	48.318510	0.001	0.0014000	0.028
SNU-668.11	SNU-668	Glucose	R_Condition	15.590379	0.001	0.0014000	0.028
SNU-668.12	SNU-668	Glucose	Reversine_Control	13.114618	0.001	0.0014000	0.028
SNU-668.13	SNU-668	Glucose	Reversine_Glucose	15.543245	0.001	0.0014000	0.028
SNU-668.14	SNU-668	K_Condition	Oxygen	45.261957	0.001	0.0014000	0.028
SNU-668.15	SNU-668	K_Condition	Phosphate	38.739638	0.001	0.0014000	0.028
SNU-668.16	SNU-668	K_Condition	R_Condition	17.480992	0.001	0.0014000	0.028
SNU-668.17	SNU-668	K_Condition	Reversine_Control	7.046331	0.002	0.0025455	0.056
SNU-668.18	SNU-668	K_Condition	Reversine_Glucose	11.525877	0.001	0.0014000	0.028
SNU-668.19	SNU-668	Oxygen	Phosphate	23.488857	0.001	0.0014000	0.028
SNU-668.20	SNU-668	Oxygen	R_Condition	2.878296	0.064	0.0663704	1.000
SNU-668.21	SNU-668	Oxygen	Reversine_Control	5.138719	0.010	0.0116667	0.280
SNU-668.22	SNU-668	Oxygen	Reversine_Glucose	4.652350	0.001	0.0014000	0.028
SNU-668.23	SNU-668	Phosphate	R_Condition	10.748819	0.001	0.0014000	0.028
SNU-668.24	SNU-668	Phosphate	Reversine_Control	11.231199	0.001	0.0014000	0.028
SNU-668.25	SNU-668	Phosphate	Reversine_Glucose	14.538138	0.001	0.0014000	0.028
SNU-668.26	SNU-668	R_Condition	Reversine_Control	4.350610	0.022	0.0236923	0.616
SNU-668.27	SNU-668	R_Condition	Reversine_Glucose	5.795666	0.002	0.0025455	0.056

	Cell_Line	Group1	Group2	F_value	P_value	P_adj_BH	P_adj_Bonf
SNU-668.28	SNU-668	Reversine_Control	Reversine_Glucose	2.347491	0.093	0.0930000	1.000
SUM-159	SUM-159	K_Condition	Oxygen	0.040959	0.866	0.8660000	0.866
SUM-159-4N	SUM-159-4N	K_Condition	Oxygen	1.342611	0.209	0.2090000	0.209

Interpretation: The pairwise PERMANOVA analysis across four cell lines (HGC-27, MDAMB231, SNU-668, SUM-159, and SUM-159-4N) revealed strong treatment-specific karyotypic divergence, particularly in HGC-27 and SNU-668, while minimal or no effects were observed in SUM-159 lines.

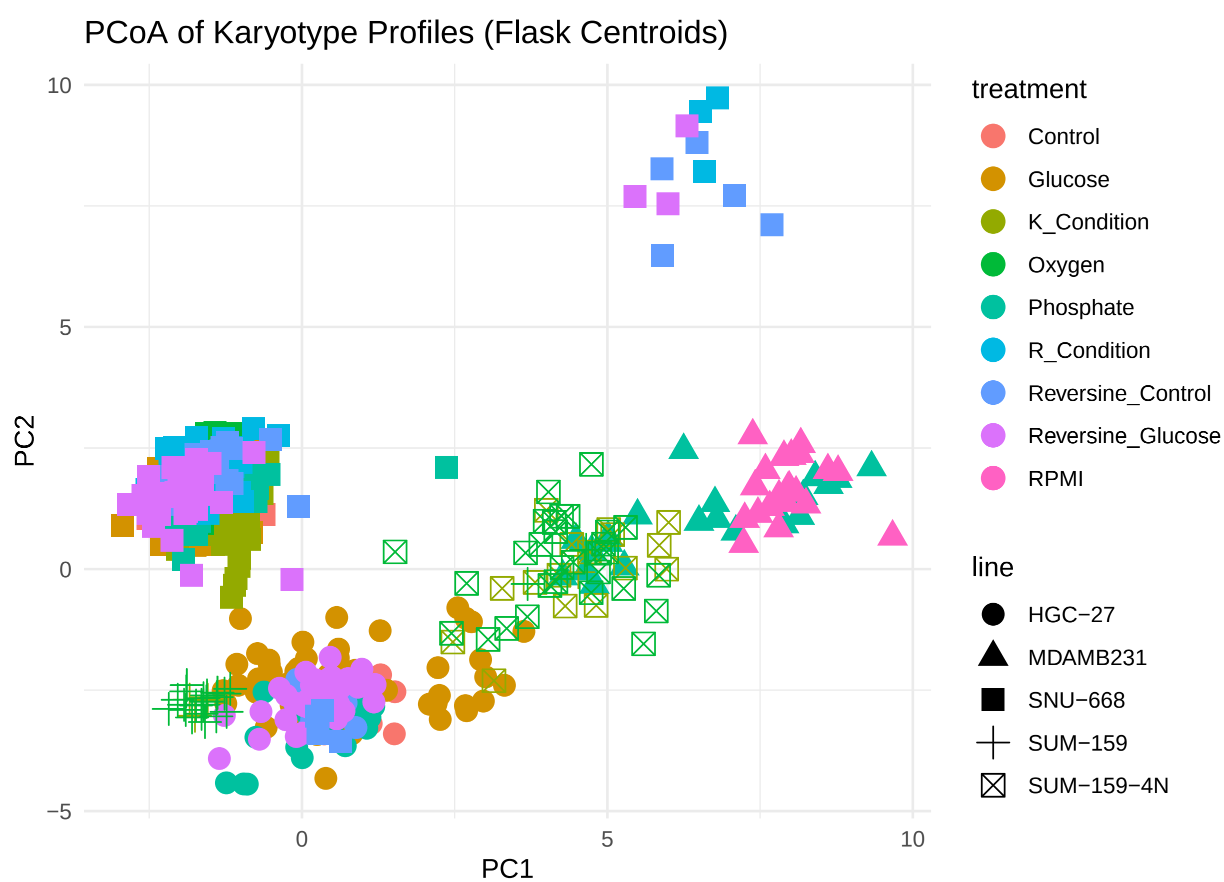
- HGC-27: All 10 pairwise comparisons were statistically significant, even after both Benjamini-Hochberg (BH) and Bonferroni corrections. This indicates robust and consistent differences in chromosomal profiles between treatments such as Control vs. Glucose, Phosphate, and both Reversine conditions. These results underscore HGC-27's high karyotypic sensitivity to environmental or drug-induced perturbations.
- MDAMB231: Only one pairwise comparison (Phosphate vs. RPMI) was significant ($p < 0.001$), suggesting a more specific and limited treatment effect on karyotype composition in this line.
- SNU-668: This line showed extensive significant pairwise differences across treatments. 27 out of 28 pairs were significant after BH adjustment (many also Bonferroni corrected), especially comparisons involving K_Condition, Oxygen, Phosphate, and Reversine_Glucose. This makes SNU-668 the most karyotypically plastic line in response to diverse conditions.
- SUM-159 & SUM-159-4N: No pairwise comparisons reached significance. This suggests that karyotype profiles remain stable in these lines across the tested treatments, indicating low chromosomal variability or insensitivity to the perturbations.

Table 4: Pairwise Comparisons with p-value < 0.005

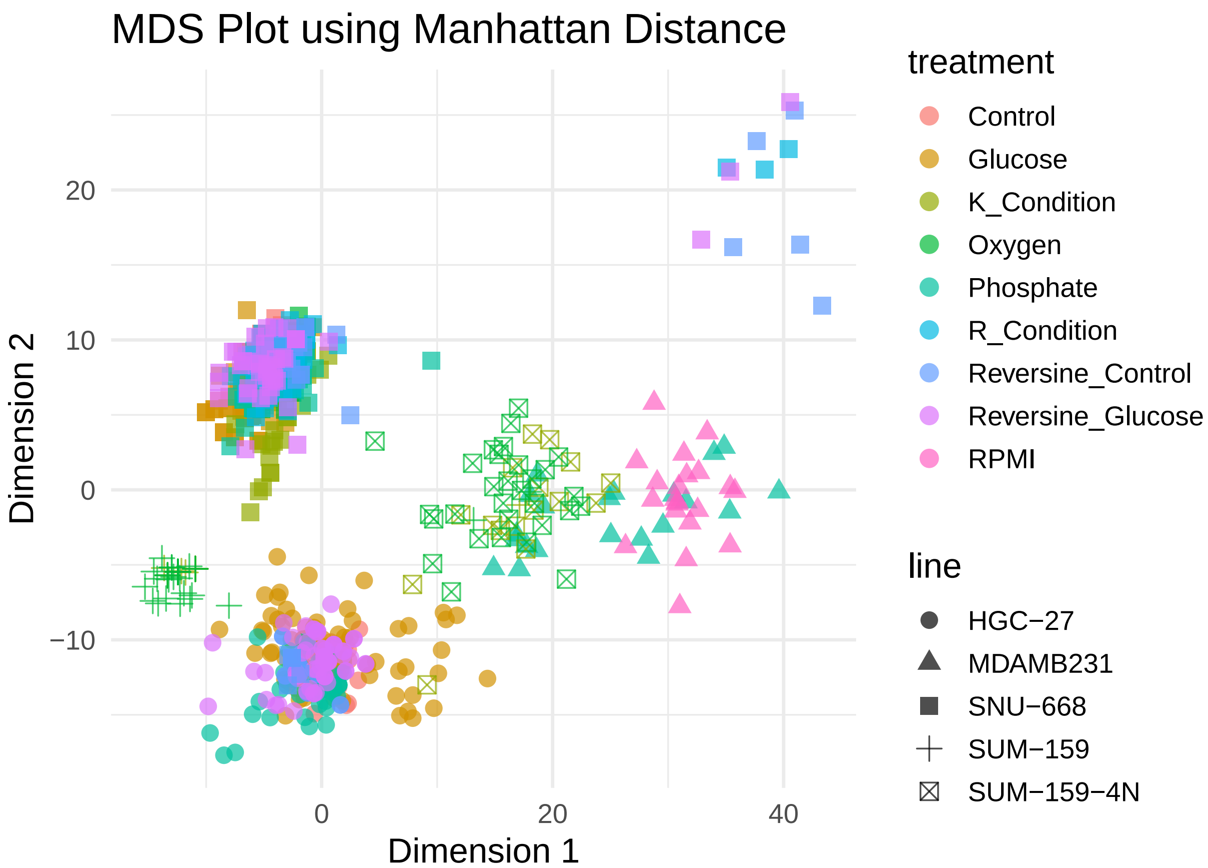
	Cell_Line	Group1	Group2	F_value	P_value	P_adj_BH	P_adj_Bonf
HGC-27.1	HGC-27	Control	Glucose	3.863941	0.003	0.0030000	0.030
HGC-27.2	HGC-27	Control	Phosphate	9.250636	0.001	0.0011111	0.010
HGC-27.3	HGC-27	Control	Reversine_Control	13.829547	0.001	0.0011111	0.010
HGC-27.4	HGC-27	Control	Reversine_Glucose	18.685173	0.001	0.0011111	0.010
HGC-27.5	HGC-27	Glucose	Phosphate	8.574273	0.001	0.0011111	0.010
HGC-27.6	HGC-27	Glucose	Reversine_Control	6.504384	0.001	0.0011111	0.010
HGC-27.7	HGC-27	Glucose	Reversine_Glucose	6.160983	0.001	0.0011111	0.010
HGC-27.8	HGC-27	Phosphate	Reversine_Control	6.683576	0.001	0.0011111	0.010
HGC-27.9	HGC-27	Phosphate	Reversine_Glucose	8.420818	0.001	0.0011111	0.010
HGC-27.10	HGC-27	Reversine_Control	Reversine_Glucose	11.607263	0.001	0.0011111	0.010
MDAMB231	MDAMB231	Phosphate	RPMI	9.834362	0.001	0.0010000	0.001
SNU-668.1	SNU-668	Control	Glucose	8.466687	0.001	0.0014000	0.028
SNU-668.2	SNU-668	Control	K_Condition	48.349747	0.001	0.0014000	0.028
SNU-668.3	SNU-668	Control	Oxygen	24.726031	0.001	0.0014000	0.028
SNU-668.4	SNU-668	Control	Phosphate	41.516836	0.001	0.0014000	0.028
SNU-668.5	SNU-668	Control	R_Condition	7.578768	0.001	0.0014000	0.028
SNU-668.6	SNU-668	Control	Reversine_Control	6.072591	0.003	0.0036522	0.084
SNU-668.8	SNU-668	Glucose	K_Condition	61.156951	0.001	0.0014000	0.028
SNU-668.9	SNU-668	Glucose	Oxygen	24.790881	0.001	0.0014000	0.028
SNU-668.10	SNU-668	Glucose	Phosphate	48.318510	0.001	0.0014000	0.028
SNU-668.11	SNU-668	Glucose	R_Condition	15.590379	0.001	0.0014000	0.028
SNU-668.12	SNU-668	Glucose	Reversine_Control	13.114618	0.001	0.0014000	0.028
SNU-668.13	SNU-668	Glucose	Reversine_Glucose	15.543245	0.001	0.0014000	0.028
SNU-668.14	SNU-668	K_Condition	Oxygen	45.261957	0.001	0.0014000	0.028
SNU-668.15	SNU-668	K_Condition	Phosphate	38.739638	0.001	0.0014000	0.028
SNU-668.16	SNU-668	K_Condition	R_Condition	17.480992	0.001	0.0014000	0.028
SNU-668.17	SNU-668	K_Condition	Reversine_Control	7.046331	0.002	0.0025455	0.056
SNU-668.18	SNU-668	K_Condition	Reversine_Glucose	11.525877	0.001	0.0014000	0.028
SNU-668.19	SNU-668	Oxygen	Phosphate	23.488857	0.001	0.0014000	0.028
SNU-668.22	SNU-668	Oxygen	Reversine_Glucose	4.652350	0.001	0.0014000	0.028

	Cell_Line	Group1	Group2	F_value	P_value	P_adj_BH	P_adj_Bonf
SNU-668.23	SNU-668	Phosphate	R_Condition	10.748819	0.001	0.0014000	0.028
SNU-668.24	SNU-668	Phosphate	Reversine_Control	11.231199	0.001	0.0014000	0.028
SNU-668.25	SNU-668	Phosphate	Reversine_Glucose	14.538138	0.001	0.0014000	0.028
SNU-668.27	SNU-668	R_Condition	Reversine_Glucose	5.795666	0.002	0.0025455	0.056

```r
# Step 7: Visualize results using PCoA (ordination plot)
pcoa <- cmdscale(dist_flask, eig = TRUE, k = 2)
pcoa_df <- data.frame(PC1 = pcoa$points[,1],
                      PC2 = pcoa$points[,2],
                      treatment = meta_flask$treatment,
                      line = meta_flask$line)

ggplot(pcoa_df, aes(x = PC1, y = PC2, color = treatment, shape = line)) +
  geom_point(size = 4) +
  theme_minimal() +
  ggtitle("PCoA of Karyotype Profiles (Flask Centroids)")
```



```r
# MDS using Manhattan Distance

# Load necessary libraries
library(vegan)
library(ggplot2)
library(dplyr)


# Step 1: Extract the chromosome features (V1 to V22)
karyo_matrix <- as.matrix(karyo_df[, paste0("V", 1:22)])

# Step 2: Compute Manhattan distance
dist_manhattan <- dist(karyo_matrix, method = "manhattan")

# Step 3: Perform MDS
mds_fit <- cmdscale(dist_manhattan, k = 2, eig = TRUE)

# Step 4: Prepare metadata
mds_df <- as.data.frame(mds_fit$points)
colnames(mds_df) <- c("Dim1", "Dim2")
mds_df$replicate_id <- karyo_df$replicate_id
mds_df$treatment <- karyo_df$treatment
mds_df$line <- karyo_df$line

# Step 5: Visualize
ggplot(mds_df, aes(x = Dim1, y = Dim2, color = treatment, shape = line)) +
  geom_point(alpha = 0.7, size = 3) +
  labs(title = "MDS Plot using Manhattan Distance",
       x = "Dimension 1", y = "Dimension 2") +
  theme_minimal(base_size = 14)
```


MDS Plot using Manhattan Distance
Dimension 2
Dimension 1
20
10
0
−10
0
20
40
treatment
Control
Glucose
K_Condition
Oxygen
Phosphate
R_Condition
Reversine_Control
Reversine_Glucose
RPMI
line
HGC−27
MDAMB231
SNU−668
SUM−159
SUM−159−4N